

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 10: Optical and Instrument Calibration

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Overview

Wavelength dependence, detector response, and the optical coupling are fit jointly with their full covariance matrix.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$\gamma(\nu) = \gamma + \gamma_c(\nu - \nu_0)/\nu_0$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$c_{read} = \lambda c_m$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$C_{opt} = \begin{pmatrix} \sigma_\gamma^2 & \cdots \\ \vdots & \sigma_\lambda^2 \end{pmatrix}$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- SGT9 optical covariance supplement
- Brovary Cross-Instrument Report
- Hall, ch. 10